ASSIGNMENT – 1

DEEP LEARNING

Name	Kamalesh T
Registration Number	192424228
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Course	Deep Learning
CO	CO1 – Mathematical Foundations and Machine Learning Principles

TASK 1 – Patient Data Representation and Normalization (5 Marks)

(a) Effect of using raw features: Age, blood pressure, glucose and cholesterol have different units and numerical

ranges. If raw values are used directly, features with larger numerical scales may dominate distance calculations and optimization. This can produce biased similarity measurements, slow or unstable gradient-based learning, and

make model training less effective.

(b) Feature matrix and min–max normalization: Consider the following illustrative patient data.

Patient Age Blood pressure Glucose Cholesterol

P1 25 120 90 180

P2 50 140 150 220

P3 75 160 210 260

The feature matrix is X = [[25, 120, 90, 180], [50, 140, 150, 220], [75, 160, 210, 260]]. Apply min–max normalization column-wise using $x' = (x - x_{min})/(x_{max} - x_{min})$.

Patient Age' BP' Glucose' Cholesterol'

P1 0.00 0.00 0.00 0.00

P2 0.50 0.50 0.50 0.50

P3 1.00 1.00 1.00 1.00

Verification: for P2 age, (50-25)/(75-25) = 25/50 = 0.5. The same calculation gives 0.5 for each other P2 feature. All normalized values lie between 0 and 1.

(c) Importance: Distance-based methods such as K-nearest neighbours use feature distances, so scaling prevents a large-range feature from dominating. Gradient-based methods benefit because similarly scaled inputs improve optimization and can make gradient descent more stable and efficient.

TASK 2 – Vector Representation and Matrix Operations (2 Marks)

(a) Let one patient's feature vector be x = [age, glucose, blood pressure, cholesterol] = [40, 150, 130, 200]. Let the

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corresponding importance-weight vector be w = [0.1, 0.4, 0.2, 0.3]. The weighted score is the dot product: $S = xw^T$

= 0.1(age) + 0.4(glucose) + 0.2(BP) + 0.3(cholesterol).

(b) S = (40×0.1) + (150×0.4) + (130×0.2) + (200×0.3) = 4 + 60 + 26 + 60 = 150. The preliminary weighted score is

150. It is an illustrative score, not a clinically validated disease probability. In practice, features should generally be

scaled and weights validated before clinical use.

TASK 3 – Probability and Conditional Probability (6 Marks)

Let D = patient has the disease, and G = patient has elevated glucose. Total patients = 1,000. Diseased = 200; non-diseased = 800. Among diseased, 160 have elevated glucose; among non-diseased, 120 have elevated glucose.

(a) Probability of elevated glucose: Total with elevated glucose = 160 + 120 = 280. Therefore, P(G) = 280/1000

= 0.28 or 28%.

(b) Probability of disease given elevated glucose: P(D|G) = P(D˙G)/P(G) = 160/280 = 0.5714 = 57.14%.

(c) Probability of elevated glucose given disease: P(G|D) = 160/200 = 0.80 or 80%.

(d) Healthcare interpretation: Conditional probability helps estimate how strongly a finding is associated with disease. In this example, 57.14% of patients with elevated glucose have the disease in this sample. This can help prioritize patients for confirmatory tests, but elevated glucose alone cannot establish a diagnosis.

TASK 4 – Bayesian Statistics for Disease Classification (4 Marks)

Given: P(D)=0.10, P(not D)=0.90, P(+|D)=0.90 and P(+|not D)=0.05.

(a) Bayes' theorem: P(D|+) = [P(+|D)P(D)] / P(+), where P(+) = P(+|D)P(D) + P(+|not D)P(not D).

P(+) = (0.90×0.10) + (0.05×0.90) = 0.09 + 0.045 = 0.135.

P(D|+) = 0.09/0.135 = 0.6667. Therefore, the probability of disease after a positive test is approximately 66.67%.

(b) The prior probability, 10%, is the disease probability before the test result. The likelihood P(+|D)=90% describes how often the test is positive among diseased patients. The posterior probability, 66.67%, is the updated

probability after considering the positive result.

(c) Bayesian reasoning combines prior knowledge with new evidence and updates the estimate as information becomes available. It is useful in healthcare because test results are uncertain and false positives or false negatives may occur.

TASK 5 – Entropy and Information Gain (3 Marks)

(a) Entropy: There are 40 high-risk and 60 low-risk records out of 100. Thus p(high)=0.4 and p(low)=0.6. H = -Spnlognpn = -[0.4logn(0.4) + 0.6logn(0.6)] = 0.971 bits (approximately).

(b) Entropy measures uncertainty or impurity in the class labels. A value near 1 bit for two classes indicates considerable uncertainty; entropy is zero when all records belong to one class.

(c) Information gain is the reduction in entropy after splitting the dataset using a feature. IG = H(parent) - weighted

average H(children). A glucose-level split with higher information gain produces purer groups and may be a useful

decision-tree feature.

TASK 6 – Numerical Computation and Distance Measurement (3 Marks)

Patient A = (0.70, 0.80, 0.60), Patient B = (0.50, 0.75, 0.90).

(a) Euclidean distance: d(A,B) = (cid:214)[(0.70-0.50)² + (0.80-0.75)² + (0.60-0.90)²] = (cid:214)[0.20² + 0.05² + (-0.30)²] =

(cid:214)[0.04 + 0.0025 + 0.09] = (cid:214)0.1325 = 0.364 approximately.

(b) A smaller distance means the patients have more similar values across the selected features. The distance 0.364 indicates a measurable difference, but whether it is clinically meaningful depends on the data and chosen features.

(c) Normalization places features on comparable scales so that a feature measured in larger units does not dominate the distance.

TASK 7 – Bias–Variance Trade-off (4 Marks)

(a) Model A has high training and validation error, indicating high bias (underfitting). Model B has very low training

error but high validation error, indicating high variance (overfitting). Model C has low and similar training and validation errors, indicating good generalization.

(b) A model with low complexity may be too simple and have high bias. Increasing complexity can reduce bias, but

excessive complexity can make the model sensitive to training data and increase variance.

(c) Improvements: For Model A, use a more expressive model, add useful features, or reduce excessive regularization. For Model B, use regularization, simplify the model, collect more representative training data, apply

cross-validation, or use early stopping.

TASK 8 – Overfitting, Underfitting and Model Capacity (4 Marks)

(a) The simple model that fails to capture disease patterns is underfitting. The moderately complex model that performs consistently shows appropriate generalization. The highly complex model that performs almost perfectly on training data but poorly on unseen patients is overfitting.

(b) Model capacity is the ability of a model to represent relationships in data. Low capacity may miss complex patterns, while higher capacity can learn more complicated relationships.

(c) Increasing capacity may initially improve performance because the model can capture meaningful patterns. Beyond a suitable level, it may also learn noise and accidental details in the training set, reducing performance on

unseen data. Validation data, regularization and suitable model selection help control this.

TASK 9 – Supervised Learning and Model Evaluation (3 Marks)

(a) This is supervised learning because historical patient records contain input features and known target labels (diseased or disease-free). The model learns the relationship between features and labels.

(b) The training set is used to fit the model and learn patterns. The testing set is kept separate and used to evaluate performance on unseen records.

(c) Accuracy: Accuracy = (correct predictions / total test predictions) × 100 = (92/100)×100 = 92%. The model

correctly classified 92 out of 100 test patients. Accuracy alone may not be sufficient if classes are imbalanced; sensitivity, specificity, precision and F1-score should also be considered.

TASK 10 – Unsupervised Learning and Patient Grouping (4 Marks)

(a) Unsupervised learning is suitable because the patient records have features but no predefined risk labels. The

method can discover structure or groups without target labels.

(b) K-means: First choose K cluster centres. Assign each patient to the nearest centre using a distance measure. Recalculate each centre as the mean of the assigned patients, then repeat assignment and update steps until the clusters stabilize.

(c) Distance measures such as Euclidean distance quantify how different two normalized patient feature vectors

are. Smaller distances indicate greater similarity in the selected measurements.

(d) Clusters may reveal groups with similar age, blood pressure, glucose and cholesterol patterns. Clinicians can

investigate these groups further and decide whether additional tests are needed. Clustering is an exploratory aid and must not replace professional diagnosis.

TASK 11 – Comparison of Supervised and Unsupervised Learning (4 Marks)

(a) Dataset A has disease labels, so supervised learning is suitable. Dataset B has no labels, so unsupervised learning is suitable for discovering hidden groups or patterns.

Aspect	Supervised learning	Unsupervised learning
Input	Features with known labels	Features without labels
Objective	Learn to predict a target	Discover structure or groups
Typical output	Predicted disease class or probability	Clusters or discovered patterns
Example	Disease classification	K-means patient grouping

(c) For predicting the disease status of a new patient, a supervised classifier trained on reliable labelled records is

the direct choice. It can output a class or estimated probability. Unlabelled records may additionally be clustered to

discover patterns and support exploration, but the final prediction should be validated against labelled outcomes.

TASK 12 – Design Challenge: Selecting and Adapting a Learning Strategy (8 Marks)

A. Impact Analysis (2 Marks)

(a) Increasing the dataset from 10,000 to 1,000,000 records increases storage, memory and computation requirements. Feature representation must remain consistent, and computation may need batching, efficient data

pipelines or scalable infrastructure. Training and evaluation may take longer, so algorithms and resource use should be selected carefully.

(b) Normalization prevents differences in feature scales from distorting distance and optimization. Missing values should be investigated and handled using suitable methods, such as median imputation for numerical features, with missingness indicators where useful. Data variability and quality matter because noisy, biased or unrepresentative records can reduce reliability and generalization.

B. Evaluation of the Original Approach (2 Marks)

(a) A single supervised approach is not sufficient for both objectives. Classification predicts disease status from labelled examples, whereas patient-group discovery is an unsupervised task that seeks structure in data without labels.

(b) Labelled learning depends on the availability and quality of target labels. If it ignores unlabelled records, potentially useful patterns and subgroups in those records may remain undiscovered. Label errors or limited labelled coverage can also restrict model performance.

C. Adaptation or Redesign (2 Marks)

(a) Use a two-part strategy: (1) a supervised classification model trained on labelled records to predict disease risk, and (2) an unsupervised clustering model applied to normalized patient features, including suitable unlabelled records, to discover patient groups. Compare cluster profiles with clinical knowledge and investigate them with healthcare professionals.

(b) Incorporate probability by producing calibrated disease-risk estimates and using Bayesian reasoning where prior information is appropriate. Use normalized feature vectors and a suitable distance measure, such as Euclidean distance, for clustering. Handle missing values before training. Control model capacity through validation and regularization to reduce overfitting.

D. Evidence-Based Defence (2 Marks)

(a) Experiments and evaluation: Compare the original classifier with the redesigned classifier using the same held-out test set and consistent preprocessing. Evaluate classification using accuracy, precision, recall/sensitivity, specificity, F1-score, ROC-AUC and calibration. Evaluate clustering using silhouette score, cluster stability and clinical interpretability. Measure training time, prediction time and memory use. Use cross-validation where appropriate and avoid data leakage.

(b) Final justification: The combined design matches each objective to a suitable learning paradigm: supervised classification for disease prediction and unsupervised clustering for patient-group discovery. Normalization and missing-value handling improve data consistency; probability supports uncertain predictions; distance supports

similarity analysis; and capacity control improves generalization. Scalable preprocessing, batching and efficient algorithms address the increase to one million records. The design should be accepted only after evaluation on representative, independent data and clinical review.

Conclusion

The healthcare risk-prediction problem requires sound mathematical preparation and an appropriate combination of learning methods. Normalization, vector operations, probability, Bayes' theorem, entropy and distance measures

support reliable analysis. Bias–variance control and separate evaluation data help assess generalization. A combined supervised and unsupervised strategy can support both disease prediction and patient-group discovery, while clinical decisions remain the responsibility of qualified healthcare professionals.